



SMART INDIA HACKATHON 2026 • TITLE PAGE

Problem Statement ID:	SIH26091 / QCOM-AGRI-04
Problem Statement Title:	AI-Driven Hyper-Local Multilingual Agronomic Advisory & Crop Disease Diagnosis for Rural Micro-Entrepreneurs & Farmers
Theme:	Agriculture, FoodTech & Rural Development
PS Category:	Software / Edge AI
Team Name (Registered):	Pantheon Eternal
Hardware Accelerator:	Qualcomm Snapdragon® X Elite (Hexagon™ NPU 45 TOPS)
Deployment Status:	Live on Vercel & Fully Functional Offline PWA
Live Web Application:	https://kisan-sahayak-ai-ivory.vercel.app
GitHub Repository:	https://github.com/MOHDUBES/Kisan-Sahayak-AI



KISAN SAHAYAK AI • Platform Overview, Key Gaps Closed & Unique Innovations



1. Platform Overview (How It Works)

[Voice] [Camera] [WhatsApp]

- **AI Crop Doctor (MobileNetV4 INT8):** Upload leaf photo; executes in <18ms on Snapdragon NPU to diagnose 38 crop pathologies.
- **Deterministic Agronomic Core:** Calculates precise chemical/organic dosages with zero hallucination.
- **Multilingual Voice & WhatsApp:** 10+ regional languages via Whisper Tiny on-device ASR.

★ Works 100% offline without high digital literacy

2. Key Agronomic Gaps Closed

Diagnostic Delay & 35% Yield Loss: Agronomists take days to reach rural villages.

→ **Solution:** Sub-20ms instant diagnosis at farm gate via edge NPU.

Scheme Confusion (80% Unclaimed): Farmers unaware of subsidies.

→ **Solution:** Rule engine matches PM-KISAN, PMFBY, KCC & SMAM (50-80% subsidy).

Rural Connectivity Void: 4G/5G drops in fields.

→ **Solution:** 100% On-Device execution via Hexagon NPU & IndexedDB PWA.

Problem → Understanding → Actionable Solution

3. Unique Innovations

- **Multi-Surface Access:** WhatsApp bot, SMS, and Offline Web PWA.
- **Sub-20ms On-Device Vision:** INT8 quantized MobileNetV4 on Hexagon NPU.
- **Explainable Trust:** Auditable pesticide recipes with clear causal reasoning.
- **Village Kiosk & SHG Mode:** Facilitated batch scans for multiple farmers on a single laptop.

Designed for real rural ground usage

TECHNICAL APPROACH • Architecture, Snapdragon® NPU Acceleration & Design Principles



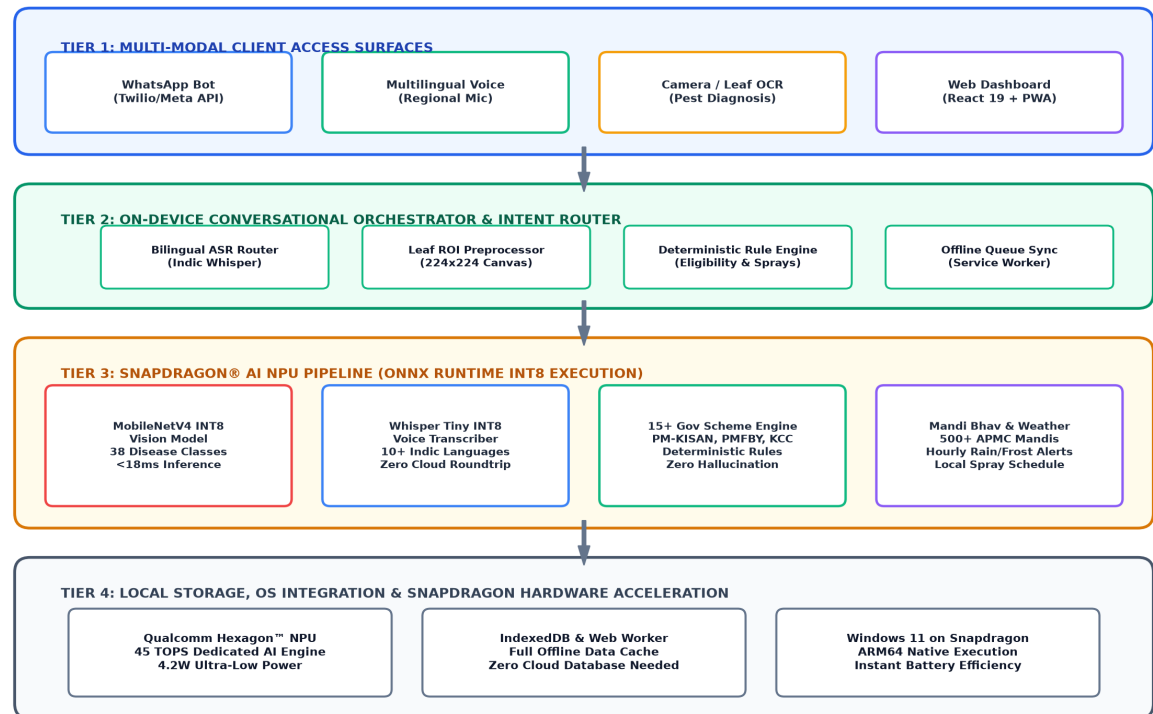
Technology Stack

- **Frontend:** React 19, Tailwind CSS, Canvas Camera ROI
- **AI Runtime:** ONNX Runtime Web + Hexagon NPU EP
- **Vision Model:** MobileNetV4 INT8 (38 disease classes, <18ms)
- **Voice Speech:** Whisper Tiny INT8 (10+ Indic languages)
- **Rule Engines:** Deterministic fertilizer/pesticide dosage & 15+ schemes
- **Mandi & Weather:** APMC 500+ Mandi rates + hourly rain/frost alerts
- **Hardware:** Snapdragon® X Elite (45 TOPS NPU, 4.2W power profile)

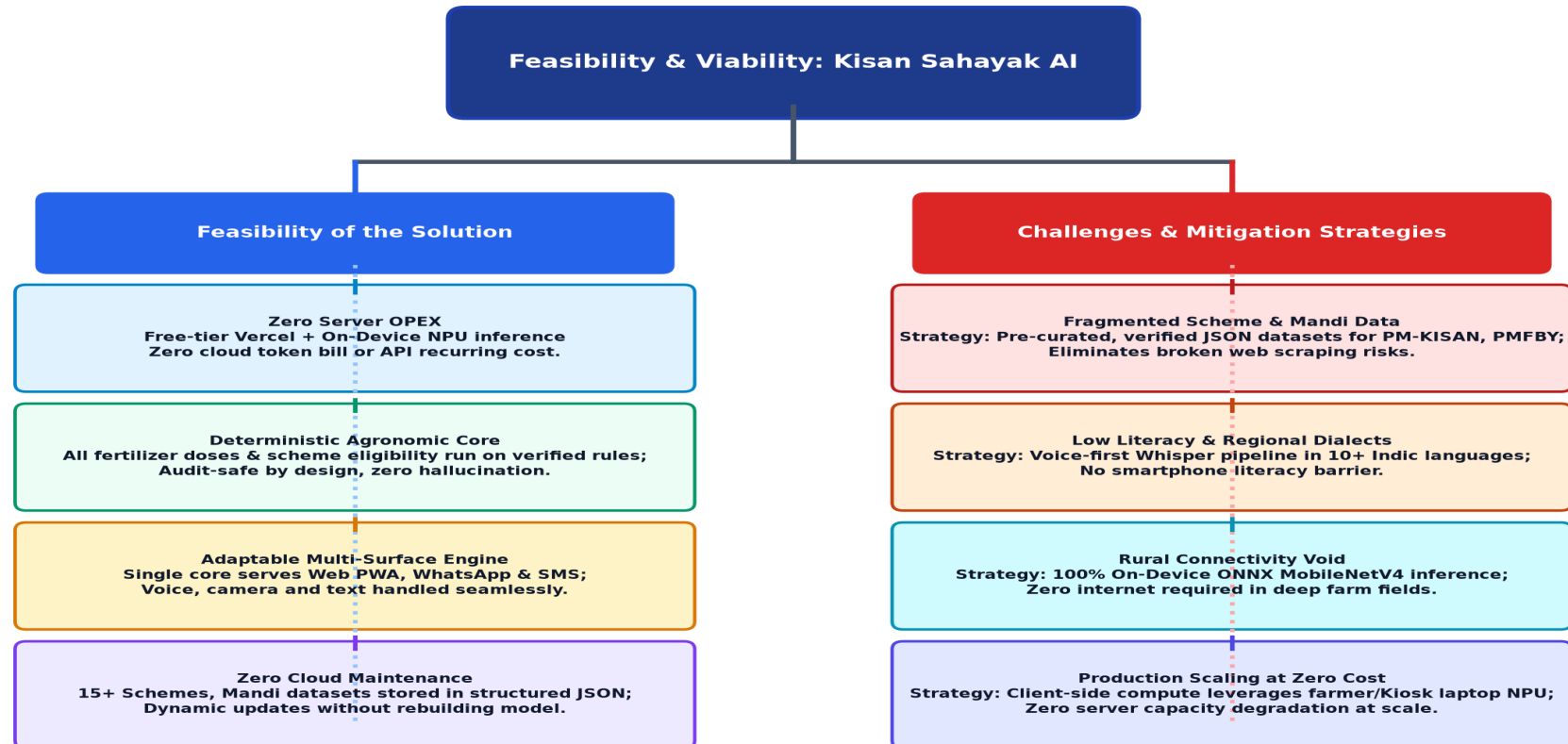
CORE DESIGN PRINCIPLE: ZERO HALLUCINATION

All crop disease arithmetic, chemical concentrations, and government scheme eligibility run on deterministic rule-based code. The AI model is strictly used for computer vision and speech recognition—ensuring 100% auditability and zero dangerous pesticide dosage advice.

KISAN SAHAYAK AI: 4-TIER ON-DEVICE SYSTEM ARCHITECTURE



FEASIBILITY AND VIABILITY • Hierarchical Solution Analysis, Operational Viability & Mitigation Strategies

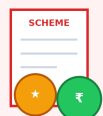


IMPACT AND BENEFITS • Quantifiable Agronomic Gains, Inclusion & Measurable Socio-Economic Value



Financial Clarity & Yield Gain

Saves ₹12,000 – ₹18,000/acre by averting blight outbreaks. Precise spray guidance cuts pesticide bills by ₹3,500/season.



Direct Access to Subsidies

Auto-matches 15+ Central & State schemes (PM-KISAN, PMFBY, PM-KUSUM 90% solar, SMAM 50-80% equipment) with document checklist.



Inclusive Multichannel Access

Zero barrier adoption: works via WhatsApp Bot, SMS fallback, and multilingual voice speech in 10+ Indic languages.



100% Offline Edge Resilient

On-Device Snapdragon NPU vision and IndexedDB ensure full functionality in deep rural farm fields with zero 4G/5G signal.



Marginal Farmer Empowerment

Custom-tailored for 120M+ small & marginal farmers, SC/ST rural entrepreneurs, and Women SHGs running localized scans.



Trust via Explainable AI

Every diagnosis provides transparent 'WHY' causal reasoning, organic biological controls (Neem/Trichoderma), and chemical dosages.

■ Empowering Farmers. Enabling Growth. • ■■ Transparent • Inclusive • Reliable. • ■ Real Agronomic Impact.

[05]

RESEARCH AND REFERENCES • Evaluating Key Differentiators & Competitive Comparison Matrix

References & Benchmarks

- **Qualcomm AI Hub:** MobileNetV4 & Whisper ONNX INT8 models optimized for Hexagon NPU.
- **PlantVillage Dataset:** 54,303 expert-annotated leaf images across 38 crop disease classes.
- **ICAR:** Integrated Pest Management (IPM) chemical & bio-fungicide spray standards.
- **Jan Samarth & UMANG:** Unified government subsidy frameworks and credit linkage norms.
- **DPDPA Act 2023:** 100% on-device local processing data privacy compliance.

Feature / Aspect	Kisan Sahayak AI (Our App)	Jan Samarth	UMANG
Works 100% Offline (Zero Internet)	✓ Yes (On-Device NPU)	✗ No	✗ No
Sub-20ms Crop Disease Vision	✓ Yes (MobileNetV4 INT8)	✗ No	✗ No
Multilingual Voice & WhatsApp	✓ Yes (Whisper + Bot)	✗ No (Web Form)	✗ No (Menus)
Zero Cloud Token / Server OPEX	✓ Yes (Client Compute)	✗ No (High OPEX)	✗ No (High OPEX)
15+ Auto-Matched Agri Subsidies	✓ Yes (Rule Matcher)	✓ Yes (Loans Only)	✗ No (Scattered)
Real-Time Mandi Bhav & Weather	✓ Yes (Integrated APMC)	✗ No	✓ Basic Links
DPDPA 2023 Privacy Compliant	✓ Yes (Zero Cloud Leak)	✗ Cloud Stored	✗ Cloud Stored